ECE 276A Project 3 Report

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***Abstract*—This paper presents the implementation of a Visual-Inertial Simultaneous Localization and Mapping (VI-SLAM) pipeline for autonomous ground vehicles operating in unknown environments. An Extended Kalman Filter (EKF) is utilized to fuse high-frequency inertial measurements with stereo visual observations. The system state is propagated using $SE(3)$ kinematics driven by IMU data, while static 3D landmarks are triangulated and continuously updated to bound the inherent drift of pure inertial dead-reckoning. To process raw image streams, a custom visual feature-extraction pipeline is developed using Shi-Tomasi corner detection and pyramidal Lucas-Kanade optical flow, reinforced by epipolar and disparity constraints for robust outlier rejection. Evaluation on real-world datasets collected by Clearpath Jackal robots demonstrates that the fully coupled joint EKF effectively constrains pose uncertainty, resulting in accurate trajectory estimation and geometrically consistent environmental mapping.**

I. INTRODUCTION

Visual-inertial simultaneous localization and mapping (VI-SLAM) is an essential capability that allows autonomous systems to navigate unknown environments by fusing internal inertial measurements with external visual information. In this project, a complete VI-SLAM pipeline is implemented to process data collected by Clearpath Jackal robots navigating outdoors on the MIT campus. The primary objective is to accurately estimate the robot's $SE(3)$ pose trajectory over time while simultaneously reconstructing a 3D map of static visual landmarks in the surrounding environment.

The general approach involves processing synchronized data streams containing linear and angular velocities from an Inertial Measurement Unit (IMU) alongside visual features extracted from stereo camera images. To achieve this, the pipeline is structured around three core components. First, a custom feature detection and tracking front-end processes raw stereo image streams using Shi-Tomasi corner detection and Lucas-Kanade optical flow to robustly extract and track visual landmarks across frames. Second, a kinematic prediction model integrates the IMU data to estimate the robot's trajectory, propagating the state and its uncertainty forward in time using Lie algebra kinematics. Finally, a joint SLAM update step utilizes an Extended Kalman Filter (EKF) backend to fuse the triangulated visual observations with the IMU predictions. By maintaining a joint dense covariance matrix, the system continuously uses the mapped visual landmarks to correct the inherent drift of pure inertial dead-reckoning.

II. PROBLEM FORMULATION

A. System State

The objective is to estimate the hidden state of the system over time, which consists of the robot's trajectory and the environment map.

- **Robot Trajectory:** The sequence of world-frame IMU poses $T_{1:T}$, where each pose $T_t := {}_WT_{I,t} \in SE(3)$.
- **Environment Map:** The static world-frame 3D coordinates of the M point landmarks, denoted as $m = \begin{bmatrix} m_1^\top & \dots & m_M^\top \end{bmatrix}^\top \in \mathbb{R}^{3M}$.

B. Motion Model

The system is driven by continuous inertial measurements. It is given control inputs $u_t := \begin{bmatrix} v_t^\top & \omega_t^\top \end{bmatrix}^\top \in \mathbb{R}^6$, consisting of the linear velocity and angular velocity expressed in the IMU body frame. The continuous-time pose kinematics are modeled as:

$$\dot{T} = T(\hat{u} + \hat{w}) \tag{1}$$

Where $w \sim \mathcal{N}(0, W)$ is the continuous-time process noise. In discrete time, this defines the probabilistic motion model $p_f(T_{t+1} \mid T_t, u_t)$, which propagates the state forward.

C. Observation Model and Feature Tracking

The system receives sequential visual observations from a stereo camera. For `dataset00` and `dataset01`, these are provided directly as pre-computed measurements $z_{1:T}$, where each $z_t \in \mathbb{R}^{4\times M}$ represents the pixel coordinates of tracked visual features. For `dataset02`, the system is given the left and right grayscale images $\mathcal{I}_{L,t}$ and $\mathcal{I}_{R,t}$. A front-end feature tracking algorithm is first applied to detect corners and compute optical flow, extracting and associating features across frames to generate the measurement matrix z_t.

Assuming known data association where landmark j corresponds to observation i at time t, the observation model mapping the spatial state to the image plane is defined as:

$$z_{t,i} = h(T_t, m_j) + v_{t,i} := K_s \pi({}_OT_I T_t^{-1} \underline{m}_j) + v_{t,i} \tag{2}$$

Where:

- $\underline{m}_j$ represents the homogeneous coordinates of landmark j.
- ${}_OT_I \in SE(3)$ is the known extrinsic transformation from the IMU to the left camera optical frame.

- K_s is the known stereo camera intrinsic calibration matrix.
- $\pi(\cdot)$ is the projection function mapping 3D coordinates to image pixels.
- $v_{t,i} \sim \mathcal{N}(0, V)$ is the Gaussian measurement noise.

This formulation defines the probabilistic observation model $p_h(z_t \mid T_t, m)$.

D. Estimation Objective

The visual-inertial SLAM problem is formulated probabilistically as a Maximum A Posteriori (MAP) estimation problem. Let the given data be $\mathcal{D} = \{z_{1:T}, u_{0:T-1}\}$ and the unknown parameters be the state trajectory and map $\theta = \{T_{0:T}, m\}$.

Using Bayes' theorem and the Markov assumption, the joint probability density function of the data and parameters is factored into the prior, the observation model, and the motion model:

$$p(\mathcal{D}, \theta) = p(T_0, m) \prod_{t=1}^{T} p_h(z_t \mid T_t, m) \prod_{t=0}^{T-1} p_f(T_{t+1} \mid T_t, u_t) \tag{3}$$

The formal objective is to find the optimal state and map parameters θ^* that maximize this joint posterior distribution:

$$\theta^* = \arg\max_{T_{0:T}, m} p(T_0, m) \prod_{t=1}^{T} p_h(z_t \mid T_t, m) \prod_{t=0}^{T-1} p_f(T_{t+1} \mid T_t, u_t) \tag{4}$$

III. Technical Approach

A. IMU Localization via EKF Prediction

The first component of the VI-SLAM pipeline involves predicting the IMU pose over time using a purely kinematic motion model driven by the IMU measurements. Because the IMU pose T_t resides in the $SE(3)$ manifold rather than a standard vector space, a standard Gaussian distribution cannot be used. Instead, the pose distribution is expressed using a deterministic nominal mean $\mu_{t|t} \in SE(3)$ and a zero-mean Gaussian perturbation $\delta\mu_{t|t} \sim \mathcal{N}(0, \Sigma_{t|t})$ on the Lie algebra $\mathfrak{se}(3)$:

$$T_t = \mu_{t|t} \exp(\hat{\delta\mu_{t|t}}) \tag{5}$$

1) Mean Prediction (Nominal Kinematics): To propagate the mean forward in time by a discretization step τ_t (derived from the timestamp differences), the continuous-time kinematics $\dot{\mu} = \mu\hat{u}$ are applied discretely. The updated nominal pose is calculated via the exponential map:

$$\mu_{t+1|t} = \mu_{t|t} \exp(\tau_t \hat{u}_t) \tag{6}$$

In this implementation, this is achieved by converting the scaled velocity vector $\tau_t u_t$ into an $SE(3)$ transformation matrix and applying it to the previous state.

2) Covariance Prediction (Perturbation Kinematics): To update the covariance $\Sigma_{t|t} \in \mathbb{R}^{6\times6}$, the propagation of the existing uncertainty through the motion model and the addition of process noise W must both be accounted for. The perturbation kinematics evolve according to:

$$\delta\mu_{t+1|t} = \exp(-\tau_t \overset{\curlywedge}{u}_t)\delta\mu_{t|t} + \sqrt{\tau_t}\omega_t \tag{7}$$

where $\omega_t \sim \mathcal{N}(0, W)$. Taking the expectation yields the EKF covariance prediction step:

$$\Sigma_{t+1|t} = \exp(-\tau_t \overset{\curlywedge}{u}_t)\Sigma_{t|t} \exp(-\tau_t \overset{\curlywedge}{\mathbf{u}}_t)^\top + \tau_t W \tag{8}$$

Programmatically, the term $\exp(-\tau_t \overset{\curlywedge}{u}_t$ acts as a linear operator on the tangent space and is implemented using the adjoint representation of the inverse motion transformation.

3) Initialization and Tuning: The filter is initialized with the IMU at the origin, setting $\mu_0 = I_{4\times4}$ and an initial covariance $\Sigma_0 = 10^{-3} * I_{6\times6}$ to reflect high initial confidence. The process noise covariance is modeled as $W = 0.01 * I_{6\times6}$.

B. Feature Detection and Tracking

For `dataset02`, a visual feature-extraction pipeline was implemented to construct the observation matrix $z_t \in \mathbb{R}^{4\times M}$. This pipeline relies on corner detection and iterative optical flow to track features both across the stereo baseline and temporally across sequential frames.

1) Feature Detection and Spatial Masking: To initialize landmarks, prominent corners are extracted from the left image $\mathcal{I}_{L,t}$ using the Shi-Tomasi corner detection algorithm. For a local window around a pixel, the structure tensor (autocorrelation matrix) S is computed using the horizontal and vertical image gradients I_x and I_y:

$$S = \sum_{(x,y)\in W} \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} \tag{9}$$

A pixel is accepted as a strong corner if the minimum eigenvalue of S exceeds a predefined quality threshold, i.e., $R = \min(\lambda_1, \lambda_2) > \tau$.

To ensure a uniform distribution of features across the image plane and prevent redundant tracking of the same physical structure, a spatial mask is maintained. Before detecting new features, circular regions (radius of 15 pixels) around currently tracked features are masked out, forcing the detector to only find corners in newly uncovered or feature-sparse regions of the image.

2) Stereo Matching via Optical Flow: For a given set of keypoints in the left image, corresponding points in the right image $\mathcal{I}_{R,t}$ are found using Pyramidal Lucas-Kanade optical flow. This method relies on the brightness constancy assumption, which posits that the pixel intensity remains constant under small displacements:

$$\mathcal{I}(x, y, t) = \mathcal{I}(x + u, y + v, t + \Delta t) \tag{10}$$

Applying a first-order Taylor series expansion yields the optical flow constraint equation $\nabla\mathcal{I}^\top \mathbf{d} + \mathcal{I}_t = 0$, where $\mathbf{d} = [u, v]^\top$ is the displacement vector. The Lucas-Kanade algorithm solves for $\mathbf{d}$ by assuming a constant flow over a local neighborhood window, yielding a least-squares solution:

$$\mathbf{d} = \left(\sum \nabla \mathcal{I} \nabla \mathcal{I}^\top\right)^{-1} \left(-\sum \nabla \mathcal{I} \mathcal{I}_t\right) \tag{11}$$

To ensure highly robust stereo correspondences, a forward-backward consistency check is enforced: points are tracked from left to right, and then backward from right to left. A correspondence is only accepted if the round-trip Euclidean error is less than 1.0 pixel.

Furthermore, two geometric constraints are applied to reject outliers:

- **Epipolar Constraint:** Because the stereo images are nearly rectified, valid matches must lie approximately on the same horizontal scanline. Matches with a vertical deviation $|\Delta y| \geq 2.0$ pixels are discarded.
- **Disparity Constraint:** Valid features must have a positive disparity $d = u_L - u_R$ within a plausible depth range $(1.0 < d < 150.0)$.

3) Temporal Tracking: Once features are identified, they are tracked temporally from the previous left image $\mathcal{I}_{L,t-1}$ to the current left image $\mathcal{I}_{L,t}$. Pyramidal Lucas-Kanade optical flow is again employed with a forward-backward consistency check, utilizing a slightly relaxed round-trip error threshold of 2.0 pixels to account for larger inter-frame motions. Features that fail this check, or track out of the image bounds, are considered lost.

4) Observation Matrix Construction: At each time step t, the temporally tracked features and any newly detected features are passed through the stereo matching block. Features that successfully yield coordinates in both the left and right frames are assigned a unique global feature ID. The pixel coordinates are then assembled into the measurement vector $z_t = [u_L, v_L, u_R, v_R]^\top$. If a previously tracked feature is lost or occluded at time t, its corresponding entries in z_t are populated with -1 to inform the EKF backend of a missing observation.

C. Landmark Mapping via EKF Update

Assuming the predicted IMU pose $T_t \in SE(3)$ is given, the second component of the pipeline focuses on estimating the static 3D coordinates of the visual landmarks $m \in \mathbb{R}^{3M}$. To keep the computational complexity manageable, each mapped landmark is treated as independent, maintaining a 3×3 covariance matrix for each rather than tracking a dense $3M \times 3M$ joint covariance matrix.

1) Feature Filtering and Initialization: At each time step, visual feature measurements $z_t \in \mathbb{R}^{4\times M}$ are received, where each observation takes the form $z = [u_L, v_L, u_R, v_R]^\top$. Before processing, missing observations (indicated by pixel coordinates set to -1) are filtered out. Furthermore, a disparity filter is applied, accepting only features with a stereo disparity $d = u_L - u_R$ between 2 and 150 pixels. This ensures the rejection of points that are infinitely far away or erroneously close, which would result in highly unstable depth estimations.

For newly observed valid features, their 3D positions in the left camera's optical frame are initialized using stereo triangulation. Given the camera focal lengths (f_x, f_y), the principal point (c_x, c_y), and the stereo baseline b, the local optical coordinates $m_{optical} = [X, Y, Z]^\top$ are recovered as follows:

$$Z = \frac{f_x b}{d}$$
$$X = \frac{(u_L - c_x)Z}{f_x}$$
$$Y = \frac{(v_y - c_y)Z}{f_y}$$

These points are then transformed from the local optical frame to the global world frame using the current IMU pose T_t and the known extrinsic calibration matrix ${}_{O_L}T_I$:

$$m_j = T_t({}_{O_L}T_I)^{-1} \underline{m}_{optical} \tag{12}$$

where $\underline{m}_{optical} = [X, Y, Z, 1]^\top$ represents the homogeneous coordinates of the triangulated point. The initial covariance $\Sigma_{0,j}$ is set to the identity matrix.

2) Observation Model and Jacobian Computation: To account for potential minor misalignments in the stereo rig, the observation model predicts visual measurements by projecting the estimated world-frame landmarks into the left and right optical frames independently.

To perform the EKF update, the Jacobian of the observation model $H_{t,j}$ evaluated at the current landmark estimate μ_j is required. The chain rule is applied to compute the derivative of the projection function with respect to the world-frame coordinates. For a given camera frame (left or right), the landmark is transformed into the camera frame as $p_C = {}_O T_I T_t^{-1} \underline{\mu}_j$. The 2×3 projection Jacobian J_{proj} evaluated at $p_C = [x, y, z]^\top$ is:

$$J_{proj} = \begin{bmatrix} f_x/z & 0 & -f_x x/z^2 \\ 0 & f_y/z & -f_y y/z^2 \end{bmatrix} \tag{13}$$

By multiplying J_{proj} by the rotational component of the world-to-camera transformation $R_{C,W} \in \mathbb{R}^{3\times3}$, the Jacobian with respect to the landmark position is obtained. The full 4×3 Jacobian $H_{t,j}$ is constructed by vertically stacking the independently computed left and right camera Jacobians:

$$H_{t,j} = \begin{bmatrix} J_{proj,L} R_{L,W} \\ J_{proj,R} R_{R,W} \end{bmatrix} \tag{14}$$

3) EKF Update and Outlier Rejection: For each previously initialized landmark observed at time t, the predicted measurement $\tilde{z}_{t,j}$ is compared to the actual visual feature observation. To ensure robustness, an outlier rejection step is implemented that discards updates where the innovation norm $||z_{t,j} - \tilde{z}_{t,j}||$ exceeds a strict threshold.

For valid observations, the EKF update is performed per-landmark using the predefined measurement noise covariance V:

$$\begin{aligned} K_{t+1} &= \Sigma_t H_{t+1}^\top (H_{t+1}\Sigma_t H_{t+1}^\top + V)^{-1} \\ \mu_{t+1} &= \mu_t + K_{t+1}(z_{t+1} - \tilde{z}_{t+1}) \\ \Sigma_{t+1} &= (I - K_{t+1}H_{t+1})\Sigma_t \end{aligned}$$

This process, derived from the standard EKF mapping update, continuously refines the spatial coordinates of the tracked features over time.

D. Visual-Inertial SLAM

The final component combines the IMU prediction step and the visual landmark mapping step into a fully coupled Visual-Inertial Simultaneous Localization and Mapping algorithm. The state encompasses both the IMU pose $T_t \in SE(3)$ and the 3D positions of M static visual landmarks $m \in \mathbb{R}^{3M}$. Consequently, the filter maintains a joint dense covariance matrix $\Sigma \in \mathbb{R}^{(6+3M)\times(6+3M)}$, initialized with high confidence for the starting pose and lower confidence for newly triangulated landmarks. The parameters used can be seen in Table I.

TABLE I: Visual-Inertial SLAM Initialization Parameters

Parameter Name	**Symbol**	**Value**
Measurement Noise Covariance	V	$5 \times I_{4\times4}$
Process Noise Covariance	W	$10^{-3} \times I_{6\times6}$
Prior Pose Covariance	$\Sigma_{T,0}$	$10^{-3} \times I_{6\times6}$
Initial Landmark Covariance	$\Sigma_{m,0}$	$1 \times I_{3\times3}$

1) Joint EKF Prediction Step: During the prediction step, the landmarks are assumed to be static; thus, only the IMU pose and its associated covariances are propagated forward in time based on the generalized velocity input u_t. The nominal pose mean is updated via the $SE(3)$ kinematics:

$$\mu_{T,t+1|t} = \mu_{T,t|t} \exp(\tau_t \hat{u}_t) \tag{15}$$

To propagate the uncertainty, the Adjoint representation of the inverse motion transformation, $F = \mathrm{Ad}_{\exp(-\tau_t u_t)} \in \mathbb{R}^{6\times6}$, is computed. The pose-pose block of the covariance matrix is updated by applying this linear operator and injecting the process noise W:

$$\Sigma_{TT,t+1|t} = F\Sigma_{TT,t|t}F^\top + \tau_t W \tag{16}$$

Because the pose and landmarks are jointly correlated, the cross-covariance terms between the pose and the mapped landmarks (Σ_{Tm} and Σ_{mT}) must also be updated to reflect the IMU's motion. The landmark-landmark covariance block (Σ_{mm}) remains unchanged.

$$\begin{aligned} \Sigma_{Tm,t+1|t} &= F\Sigma_{Tm,t|t} \\ \Sigma_{mT,t+1|t} &= \Sigma_{Tm,t+1|t}^\top \end{aligned}$$

2) Joint Observation Jacobian Computation: When new visual measurements z_{t+1} are received, the full observation Jacobian $H \in \mathbb{R}^{4N_t\times(6+3M)}$ is constructed for the N_t currently tracked landmarks. The Jacobian consists of two main blocks for each observation: the derivative with respect to the IMU pose perturbation (H_T) and the derivative with respect to the landmark position (H_m).

The landmark Jacobian $H_m \in \mathbb{R}^{4N_t\times3M}$ is computed using the chain rule on the stereo projection function, exactly as defined in the visual mapping step.

The pose Jacobian $H_T \in \mathbb{R}^{4N_t\times6}$ evaluates how a perturbation in the IMU pose affects the projected pixel coordinates. For a given landmark m_j, it is computed using the circle-dot operator $\odot$, which maps homogeneous coordinates to a 4×6 matrix to resolve the Lie algebra derivative. For a generic optical frame O (left or right), the block Jacobian is:

$$H_{T,j} = -J_{proj\,O}T_I(\mu_{T,t+1|t}^{-1}\underline{m}_j)^{\odot} \tag{17}$$

The full Jacobian H is formed by densely populating the first 6 columns with H_T and vectorizing the insertion of H_m into the corresponding 3×3 column blocks allocated for each tracked landmark.

3) Joint State and Covariance Update: Prior to the update, an outlier gating step filters out observations where the pixel error norm $||z_{obs} - z_{pred}||$ exceeds a threshold, preventing erroneous feature tracks from corrupting the joint state.

For the valid observations, the joint EKF update calculates the Kalman gain $K \in \mathbb{R}^{(6+3M)\times4N_t}$:

$$K = \Sigma_{t+1|t}H^\top(H\Sigma_{t+1|t}H^\top + I \otimes V)^{-1} \tag{18}$$

The state vector is corrected by scaling the innovation $(z - \tilde{z})$ by the Kalman gain. The correction vector is partitioned into a pose correction $\delta\mu_T \in \mathbb{R}^6$ and landmark corrections $\delta\mu_m \in \mathbb{R}^{3M}$. The IMU pose is updated via the $SE(3)$ exponential map, while the static landmarks are updated via standard vector addition:

$$\begin{aligned} \mu_{T,t+1|t+1} &= \mu_{T,t+1|t}\exp(\delta\mu_T^\wedge) \\ \mu_{m,t+1|t+1} &= \mu_{m,t+1|t} + \delta\mu_m \end{aligned}$$

Finally, the dense joint covariance matrix is updated. To optimize computational efficiency and avoid redundant multiplications with the massive H matrix, the cross-covariance component $\Sigma H^\top$ is cached and reused:

$$\Sigma_{t+1|t+1} = \Sigma_{t+1|t} - K(H\Sigma_{t+1|t}) \tag{19}$$

A symmetrization step $\Sigma = (\Sigma + \Sigma^\top)/2$ is applied to mitigate floating-point drift over time.

IV. RESULTS

The pipeline was evaluated on all three datasets, but only the `dataset00` results are presented here for brevity. The other results can be found in Section V.

A. IMU Localization via EKF Prediction

As expected for a pure dead-reckoning approach, the estimated $SE(3)$ trajectory demonstrates smooth local continuous motion but suffers from significant accumulated drift over time as seen in Figure 1. Because the kinematic model integrates linear and angular velocities to update the pose, any inherent sensor noise, unmodeled biases, or slight calibration errors are compounded at each time step. The process noise covariance W allows the filter's uncertainty bounds to grow realistically, but without the visual measurements to provide corrections, the trajectory inevitably diverges from the true path.

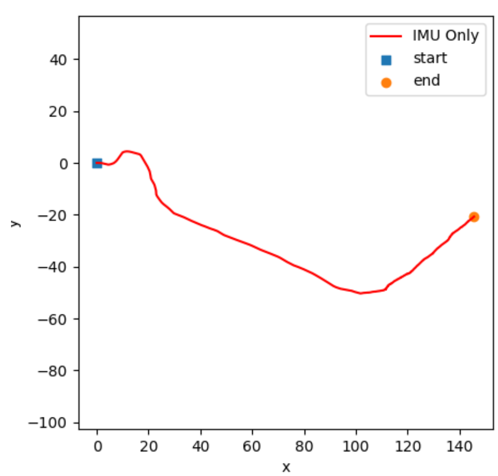


Fig. 1: Estimated IMU trajectory over time for `dataset00` using only the kinematic prediction step.

This baseline highlights the necessity of incorporating visual landmark observations to constrain the unbounded drift of the IMU odometry.

B. Feature Detection and Tracking

The visual-inertial SLAM pipeline was evaluated on `dataset02`, which provided raw stereo image streams rather than pre-computed feature correspondences. This required the full integration of the custom visual feature-extraction with the EKF backend.

1) Feature Extraction/Tracking Performance: The feature tracking front-end successfully extracted and maintained a robust set of visual landmarks across the dataset. By utilizing Shi-Tomasi corner detection combined with spatial masking, the algorithm ensured a uniform distribution of tracked features across the image plane, preventing the EKF from over-relying on a single clustered spatial region.

The bidirectional Lucas-Kanade optical flow, coupled with epipolar and disparity constraints, served as a highly aggressive outlier rejection mechanism. As seen in the feature tracking statistics, while the strict forward-backward consistency checks naturally cull features that become occluded or distorted by camera rotation, they ensure that the surviving

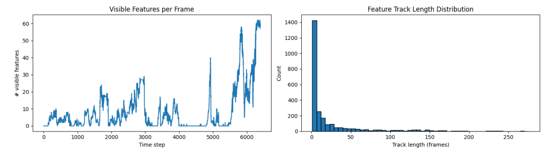


Fig. 2: *Left:* The number of actively tracked, stereo-matched features at each time step. *Right:* The distribution of feature track lengths. The strict optical flow consistency checks ensure only high-quality features are retained.

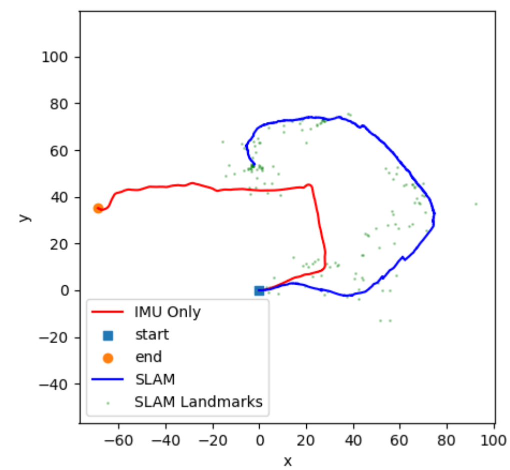


Fig. 3: The resulting Visual-Inertial SLAM trajectory for `dataset02`, utilizing the custom-tracked visual features.

tracks are of exceptional geometric quality. The tracking distribution indicates that while many features are tracked for short bursts (often due to moving out of frame during turns), a stable core of features is consistently maintained for longer durations, which is critical for continuous EKF pose correction.

2) Visual-Inertial SLAM Trajectory: The ultimate validation of the custom front-end is its performance within the joint SLAM update step. The generated feature observation matrix z_t was fed directly into the EKF, mirroring the procedure used for the pre-processed datasets.

The results demonstrate that the custom-tracked features successfully bounded the accumulated drift of the IMU kinematic prediction. During periods of pure IMU integration, the process noise allows the state covariance to grow. However, as the custom visual features are continually observed and updated, the Kalman gain effectively applies corrections to the $SE(3)$ pose. The resulting trajectory is smooth, geometrically consistent, and successfully avoids the massive heading divergence that plagues the IMU-only baseline. This confirms that the front-end not only tracks pixels accurately but provides geometrically sound 3D constraints that the EKF can reliably map and utilize for localization.

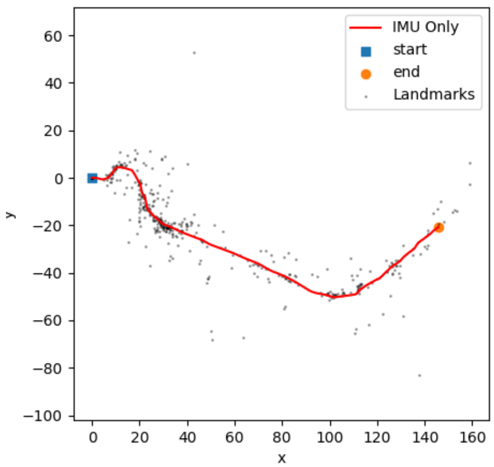


Fig. 4: Estimated map of visual landmarks from `dataset00` plotted alongside the IMU trajectory. The depth and disparity filtering successfully isolates high-confidence, static structures while rejecting distant noise and tracking anomalies.

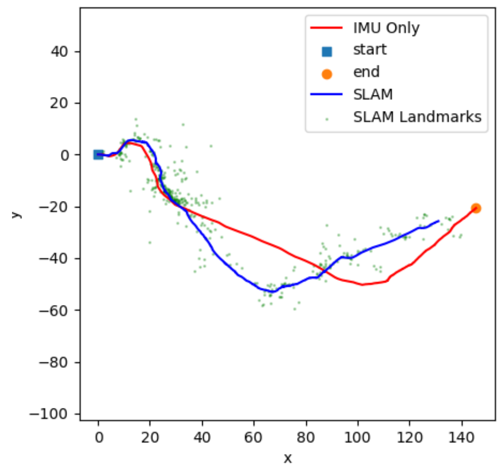


Fig. 5: Comparison of the IMU-only trajectory (red) and the joint VI-SLAM trajectory (blue) alongside mapped 3D landmarks (green).

C. Landmark Mapping via EKF Update

The landmark mapping component was evaluated by utilizing the IMU pose estimates to triangulate and update the static 3D coordinates of the observed visual features as seen in Figure 4.

A critical factor in achieving a stable and numerically sound map was the implementation of strict disparity and min/max depth filtering. In stereo vision, a landmark's depth Z is inversely proportional to the pixel disparity d between the left and right images. As disparity approaches zero, the triangulated depth approaches infinity. Consequently, for very distant features, even sub-pixel measurement noise results in massive spatial uncertainty. By enforcing a strict disparity window (e.g. $d \in [2, 150]$ pixels) and corresponding depth bounds (e.g. $Z \in [0.1, 100]$ meters), the algorithm successfully rejected ill-conditioned points.Without this filtering, these points would initialize with highly skewed covariances leading to numerical instability during the EKF update step and generating extreme spatial outliers. Furthermore, the innovation gating step successfully rejected transient feature tracking errors and dynamic objects.

It is worth noting that while the x and y coordinates of the landmarks converged cleanly, the estimation of the z coordinates was notably noisier. This behavior is expected because the vehicle primarily navigates on a flat plane without moving sufficiently along the z-axis.

D. Visual-Inertial SLAM

The fully coupled VI-SLAM algorithm combined the $SE(3)$ kinematic prediction model with the dense landmark observation updates. The results clearly demonstrate the critical role of visual updates in bounding the inherent drift of pure inertial dead-reckoning.

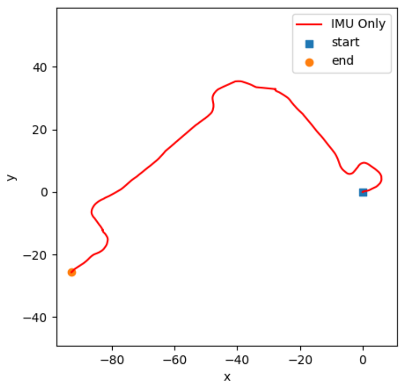
As shown in Figure 5, the IMU-only trajectory begins to suffer from accumulated drift shortly after the initial turns. Without external corrections, slight biases in the angular velocity measurements compound resulting in a heading error.

In stark contrast, the VI-SLAM trajectory maintains a precise path. By maintaining a joint dense covariance matrix, the EKF successfully captures the correlations between the vehicle's pose and the static landmark positions. During each update step, the difference between the predicted landmark pixel coordinates and the actual stereo observations generates an innovation vector. The Kalman gain applies this innovation to correct both the landmark positions and the IMU pose simultaneously.

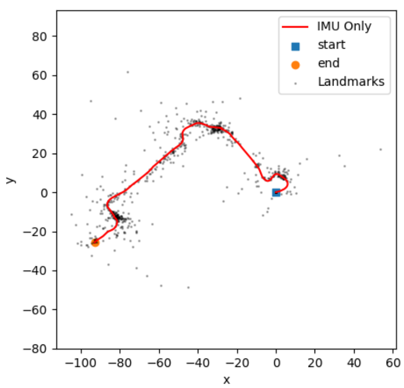
The mapped landmarks can be seen clustered tightly along the blue trajectory. As the robot navigates the curve down toward $y \approx -60$, these continuously tracked features prevent the state uncertainty from growing unbounded. The iterative cycle of high-frequency IMU predictions followed by visual corrections effectively pulls the $SE(3)$ pose back to reality resulting in a highly accurate reconstruction of the vehicle's true trajectory.

V. Appendix

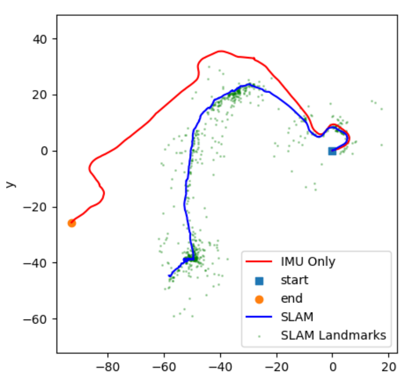
A. Dataset01 Results



IMU Only (Dataset 01)



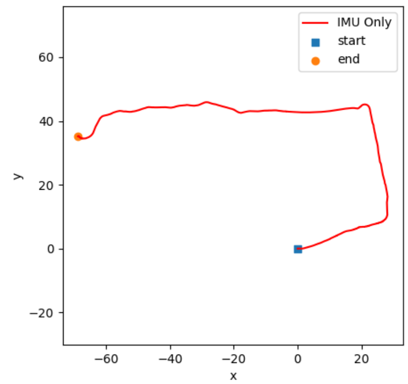
Landmark Mapping



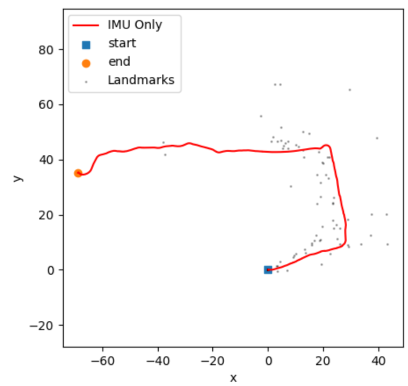
Visual-Inertial SLAM

Fig. 6: Trajectory and mapping results for Dataset 01. Similar to the primary results, the fully coupled EKF successfully bounds the accumulated drift of the IMU.

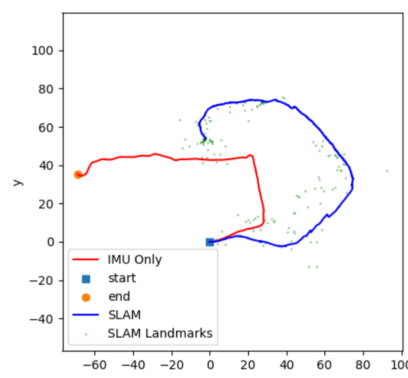
B. Dataset02 Results



IMU Only (Dataset 02)



Landmark Mapping



Visual-Inertial SLAM

Fig. 7: Trajectory and mapping results for Dataset 02. The visual features for this dataset were manually extracted and tracked across frames using Shi-Tomasi corner detection and Lucas-Kanade optical flow.